

Feedback Coding for Information-based Control: Operating Near the Data-Rate Limit

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Abstract

The paper begins with a restatement and proof of a tight bound on the data capacity a feedback channel must provide in order to stabilize a right half-plane pole. This sets the stage for a treatment of some of the issues arising in the source coding of signals between controllers and plants in systems where feedback loops are closed using bandwidth limited communication links. In particular, we shall be interested in the qualitative dynamics of LTI feedback systems operating near the data-rate limit.

1 Introduction

In a number of new technological settings we find feedback control designs are constrained by data rate limitations in the communication channels between sensors, controller, and actuators. Such constraints are a consideration, for instance, in current and future generations of MEMS arrays in which there can be as many as 10^4 to 10^6 actuators or sensors on a single chip, and on a larger physical scale they are of central interest in closing feedback loops through wireless links such as BluetoothTM or IEEE 802.11(b) and using network protocols such as CAN, [8]. Work is only just now beginning on the switching and encoding strategies that will be needed to produce stable closed-loop dynamics across a broad spectrum of applications. We are finding that many of the bandwidth assignment issues that are predominant in managing the traffic in modern communications networks are also present in some form in networked actuator arrays.

This paper will discuss control using feedback channels with severe communication constraints. The starting point for our discussion is a data rate inequality which bounds the minimum rate at which information must be transmitted from the controller to the plant in terms of the magnitude of open-loop poles lying in the right-half plane. We consider digital control implementations involving temporally uniform sampling and quantized feedback. Taking care to satisfy the data rate inequality, we study trade-offs between sampling at high temporal frequencies with coarse (spatial) quantization of the control signals and sampling at lower frequencies, but transmitting more finely quantized values of the feedback signal. Results to date suggest that as long as the data rate inequality is satisfied, there is an exact tradeoff between the coarseness of temporal and spatial quantizations. (In terms of symbols, we can use quantized feedback to obtain bounded trajectories

in a system which is open-loop unstable by transmitting n bits of data every h units of time, or we can obtain bounded trajectories for the same system by transmitting $2n$ bits of data every $h/2$ units of time over the same feedback link.)

Issues of the type discussed are motivated by several pieces of work in the recent literature. The data-rate bound of Section 2 below was presented in the workshop notes of the 1999 ARO Workshop on Smart Structures ([3]). It is also discussed in [4]. An earlier report ([12]) presented results on the optimal LQG control of LTI systems with noisy feedback channels. Incorporating a "cost of information" with the usual cost of control, it was shown in the scalar case that the average optimal cost would be finite only if the data-rate inequality was strictly satisfied. A result similar to that of Section 2 below has appeared in work by Nair and Evans ([9]), but in this case the bound was established in an entirely different way using methods of stochastic stability and asymptotic quantization theory. Extension of this result to the case of multi-dimensional LTI systems has also been recently reported by Nair and Evans ([10]). A similar bound appears in more recent work of Tatikonda and Mitter ([13]), again applying to the case of multi-dimensional LTI systems. It seems fair to say that all of this work has been inspired by results of Wong and Brockett ([14]) which treated such data-rate bounds in terms of some explicit encoding schemes. This work has been seminal in suggesting important connections between communication and control. Explicit encoding was also treated in [4] in the case of feedback systems operating near the theoretically minimum possible data rate. Even for the coarsest possible quantization of the control inputs (binary feedback), it was shown that bounded responses in open-loop-unstable linear systems could be produced. Nevertheless, for systems operating near the theoretical minimum data rate, it was shown that there could be unacceptably large sensitivity to initial conditions in the closed-loop dynamics—as well as other forms of non-robustness. The present paper represents a continuation of [4]. Recognizing that for a given feedback loop data rate we can trade off coarseness in temporal and "spatial" quantization, we explore both qualitative and quantitative aspects of decreasing the sampling rate while simultaneously passing to a more finely quantized set of control inputs.

The paper is organized as follows. To set the stage, the data rate inequality is reviewed in Section 2. First announced in [3], and extended in various ways since then ([4], [9], [10], and [13]), the inequality provides an absolute, sharp lower bound on the minimum communication bandwidth required to transmit feedback signals in order to have stable closed-loop performance of a linear time invariant system which is open-loop unstable. Although the data rate inequality is now well known in some sense, we shall present the proof as discussed in [3] because it is the most elementary proof we are aware of, and it provides good intuition into the connection between achievable bit-rate and stability. Section 3 describes the way stability breaks down at the data-rate limit in the scalar

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case. The results are quite explicit and suggest features which we expect to be present in systems of higher dimension. In Section 4, we discuss the concept of control *attention* and how so-called “binary” realizations of classical feedback designs “pay attention” to different unstable modes in higher order systems. In Section 5, we treat the qualitative dynamics of systems of dimension ≥ 2 operating near the data-rate limit. Section 6 presents our conclusions and presents some thoughts on future research.

2 Digital control of LTI systems with uniform sampling rate—the data rate bound in the scalar case

Classical feedback control theory is aimed at understanding design principles for integrating sensors and actuators for controlling a physical system in a way that it will perform its prescribed tasks efficiently and reliably. The basic principles are very simple as illustrated in Figure 1. Information about the state of the system of interest (called the plant G) is provided by sensor data. The control system (represented by the box labeled H together with the dotted interconnection) processes signals from sensor outputs, compares these values with specified operating goals, and “closes the loop” by sending appropriate actuator signals to keep the system operating as near as possible to its operating goals. There is today a vast array of mathematically sophisticated tools for design and analysis of the controller H . The tools apply to both continuous and discrete-time systems, and for many applications, they have become an essential aid for implementations of feedback control mediated by digital microprocessors. As we shall indicate below, however, feedback laws designed using classical methods may produce unexpected dynamical effects in there are significant data-rate limits on the feedback channels.

In this paper, we consider an explicit realization of the feedback system in Figure 1 of the form

$$\dot{x} = Ax + bu, \quad (1)$$

where (A, b) is a controllable pair. The case in which the control u is a scalar is the most interesting, and considering this case avoids extraneous complexity. It will also be useful to assume we have adopted the representation

$$A = \begin{pmatrix} a_1 & 0 & \dots & 0 \\ 0 & a_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_n \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix},$$

wherein the controllability condition is equivalent to assuming the diagonal entries of A are distinct. We have chosen to examine systems in this form because it makes our conclusions most transparent. While the assumption of controllability is essential, the normal form representation of the system is not.

The case we shall treat involves digital control with a uniform sampling interval h . We assume control actuation is of the sample-and-hold type, so that control inputs to the system are constant over each sampling interval. The state of the plant evolves in discrete time according to

$$x(j+1) = Fx(j) + \Gamma u(j), \quad (2)$$

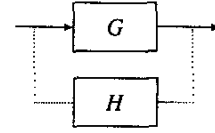


Figure 1: A feedback system. G is a rational function plant, and H is the controller—possibly having digital components.

where

$$F = e^{Ah}, \quad \text{and} \quad \Gamma = A^{-1}(e^{Ah} - I)b.$$

We adopt a very simple abstraction of a control system having a computer and data-rate limited communication links in the loop. A sensor reading of the (typically analog) value $x(k)$ is digitized and stored in a computer register. Using whatever control algorithm has been implemented, the computer determines a value of the control input $u(k)$ (from a finite set U of admissible control values), and this in turn is converted to an analog value (e.g. a voltage or a current) which is then fed back to the plant. For many applications, eight or fewer bits suffice to specify the values of the control function $u(\cdot)$, and hence the cardinality of the set U of admissible control values is typically ≤ 256 . Whatever the cardinality, $|U|$, of the set of admissible controls, the number N of bits needed to uniquely encode each control value is such that $|U| \leq 2^N$. The inequality may be strict since some encoding schemes will use a certain number of bits for purposes other than identifying control values. In recent experimental work using BluetoothTM wireless feedback links, for instance, the protocol allowed only 136 bits of data in each 366 DM1 data packet, with the remainder of packet allocated to access codes, addresses, etc. ([11]) It is important to remark that the details of the operation of system components such as computer processors, signal encoders and decoders, etc. are unimportant, and our results are only concerned with the qualitative dynamics of the closed-loop systems operating under the data-rate limits imposed by the operation of these components.

Classical notions of asymptotic stability must be modified in order to discuss control designs with quantized control values. This was noted in Wong and Brockett ([14]) where we were introduced to the notion of *containability* as a digital control concept corresponding to *asymptotic stability*. For the purposes of the present paper, we adopt a more primitive notion of stability:

Definition 1 A linear control system (1) with admissible control set U is *boundable* if there exists a bounded set S and an open set $M \subseteq S$ such that for each $x_0 \in M$ there is a control sequence $\mathcal{U} = \{u(k) \in U : k = 1, 2, \dots\}$ such that the trajectory defined by (1) with $x(0) = x_0$ and control input sequence $\mathcal{U}$ remains in S for all time.

Boundability is a very weak notion of stability, and any system which is not boundable cannot be containable in the sense of Wong and Brockett ([14]).

Example 1 *The case of binary control.* Suppose the set U of admissible control values has two elements. These correspond to

what we assume are precisely two different possible actuator commands. While these values might be different voltages or currents or other physical quantities of constant magnitude, there is no loss of generality in the model in assuming that $U = \{-1, 1\}$, since this may always be brought about for a system of the form (1) by an affine change of coordinates. To make the discussion of boundability interesting, we assume that (2) has at least one open-loop unstable mode. I.e. there is at least one element α_i on the diagonal of the (diagonal) matrix F such that $\alpha_i > 1$. In its most general form, a feedback law is defined by a selection function $f: \mathbb{R}^n \rightarrow U$ which for each $x \in \mathbb{R}^n$ chooses a control value so that the closed-loop evolution of (2) is prescribed by the formula

$$x(j+1) = Fx(j) + \Gamma f(x(j)). \quad (3)$$

Selection functions related to classical constant gain feedback designs will be treated in Sections 4 and 5.

Definition 2 We shall call a selection function f *regular* if $\{f^{-1}(u_j) : u_j \in U\}$ partitions $\mathbb{R}^n$ into exactly N connected components, each of which is homeomorphic to a closed or open half-space, where N is the cardinality $|U|$ of U .

Any digital implementation of a classical feedback design will give rise to a regular selection function. The main result of the section is the following:

Theorem 1 Consider a first order system $G(s) = \frac{b}{s-a}$ whose sampled realization (2) specializes to

$$x(j+1) = \alpha x(j) + \beta u(j), \quad (4)$$

where

$$\alpha = e^{ah}, \text{ and } \beta = \frac{b}{a}(e^{ah} - 1).$$

We may choose a sampling interval h , a finite set U of admissible control values such that $|U| > e^{ah}$, and a regular selection function such that the resulting feedback control law renders (4) boundable if and only if the channel capacity R satisfies

$$R \geq a \log_2 e. \quad (5)$$

Before presenting the proof, several remarks are in order.

Remark 1 Remarks on the channel capacity conditions. (i) Theorem 1 was presented in [3]. The proof, which we shall include here, is constructive, geometric, and uses only elementary ideas in the spirit of the seminal work of Wong and Brockett [14]. Under the assumption of the theorem, we carry out an explicit construction of the set U of control values together with a feedback law which renders a compact set invariant under motions of the system (4). (ii) Although the result of the theorem is quite sharp and explicit, it may actually represent only a lower bound on the amount of channel capacity one would like to have to implement feedback control of a physical system. To address the myriad standard control design issues such as rise time, gain and phase margins, etc., we shall need to have the capacity to transmit a great deal more data through the channel.

Remark 2 The case of binary control continued. Again consider the binary control law with $U = \{-1, 1\}$ and closed loop system (3) of the previous example. It is not difficult to see that (4) is boundable with selection function $f: \mathbb{R} \rightarrow \{-1, 1\}$ given by $\{f(x) = \text{sgn}(x), x \neq 0, f(0) = 0\}$ if and only if $\alpha \leq 2$. Taking logarithms allows us to rewrite this inequality as $a \log_2 e \leq 1/h$. Clearly we can transmit the required 1-bit of information in each length h sampling interval if and only if $1/h \leq R$.

Remark 3 Control information versus control authority. An interesting aspect of both the binary control example and the main theorem is that the conditions for boundability and the existence of invariant intervals do not depend on β . They depend only on the amount of information that can be communicated between the controller and plant in one unit of time.

Proof of Theorem 1. Consider the discretized system (4) and a finite (N -element) set of admissible control values U . Let $u_{\min}$ and $u_{\max}$ be respectively the smallest and largest members of this set. We may assume, without loss of generality, that U contains both positive and negative values, and that $u_{\min} = -u_{\max}$. Since we are considering only regular selection functions, there is a partition of the state-space $-\infty < x_1 < \dots < x_{N-1} < \infty$ such that $f(x)$ takes different values in U depending on which of the subintervals $(-\infty, x_1], (x_1, x_2], \dots, (x_{N-1}, \infty)$ x lies in. There is no loss of generality in assuming $f(x) = u_{\max}$ if $-\infty < x \leq x_1$ and $f(x) = u_{\min}$ if $x_{N-1} < x < \infty$. Label the remaining control values $u_1, \dots, u_{N-2}$ so that we may write

$$U = \{u_{\min}, u_{\max}, u_1, \dots, u_{N-2}\}.$$

We may assume the numbering has been done so that the selection function can be explicitly written

$$f(x) = \begin{cases} u_{\max} & \text{if } -\infty < x \leq x_1 \\ u_1 & \text{if } x_1 < x \leq x_2 \\ \vdots & \\ u_{N-2} & \text{if } x_{N-2} < x \leq x_{N-1} \\ u_{\min} & \text{if } x_{N-1} < x < \infty \end{cases}$$

so that the closed-loop transition function is given explicitly by

$$x(k+1) = \begin{cases} \alpha x(k) + \beta u_{\max} & \text{if } -\infty < x \leq x_1 \\ \alpha x(k) + \beta u_1 & \text{if } x_1 < x \leq x_2 \\ \vdots & \\ \alpha x(k) + \beta u_{N-2} & \text{if } x_{N-2} < x \leq x_{N-1} \\ \alpha x(k) + \beta u_{\min} & \text{if } x_{N-1} < x < \infty. \end{cases} \quad (6)$$

The graph of a typical transition function constructed in this way is depicted in Figure 2 (where $x_1 = r_1$ and $x_{N-1} = r_4$).

A simple calculation shows that the transition mapping $x(k+1) = \alpha x(k) + \beta u_{\max}$ has a fixed point at $-\beta u_{\max}/(\alpha - 1)$, and similarly $x(k+1) = \alpha x(k) + \beta u_{\min}$ has a fixed point at $-\beta u_{\min}/(\alpha - 1) = \beta u_{\max}/(\alpha - 1)$. (In Figure 2, $\beta u_{\max}/(\alpha - 1) = 1$.)

One can next show that the largest possible invariant interval for the transition function (6) is $[-\beta u_{\max}/(\alpha - 1), \beta u_{\max}/(\alpha - 1)]$. Obviously the subinterval will be invariant if and only if no point inside this interval is ever mapped outside the interval under the transition (6). A typical case in which this interval is invariant under (6) is depicted in Figure 6.

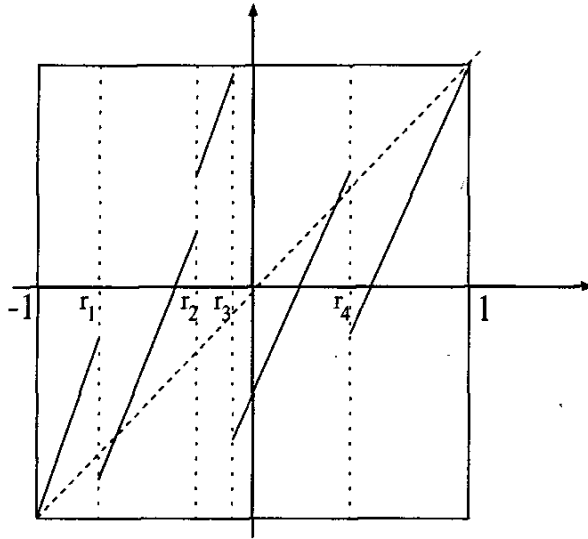


Figure 2: Feedback defined by a regular selection function assigns one of N ($= 5$ in this case) control values to each subinterval in the partition of the state space. The closed-loop transition function consists of several linear segments each having the same slope.

All lines in the graph of (6) have slope $\alpha = e^{ah}$. Denote $x_0 = -\beta u_{\max}/(\alpha - 1)$ and $x_N = \beta u_{\max}/(\alpha - 1)$. Let ℓ = length of the largest subinterval in the partition $x_0 < x_1 < \dots < x_N$, and note that the slope α must satisfy $\alpha \ell \leq 2\beta u_{\max}/(\alpha - 1)$. To make α as large as possible, we choose ℓ to be as small as possible—i.e. $\ell = (2/N) \cdot \beta u_{\max}/(\alpha - 1)$. In this case, we must have $\alpha \leq N$. Hence we have $e^{ah} \leq N$. Taking logarithms of both sides and dividing by h , we have

$$a \log_2 e \leq \frac{\log_2 N}{h}.$$

Now $\lceil \log_2 N \rceil$ is the minimum number of bits required to encode the N possible values of the control in U . In order to be able to transmit an arbitrary control value every h units of time, it must happen that the data-rate R satisfies $R \geq \log_2 N/h$. Combining these inequalities proves the necessary condition $R \geq a \log_2 e$.

The proof of sufficiency is straightforward. Given $R \geq a \log_2 e$, choose $h > 0$ and a positive integer N such that $a \log_2 e \leq \frac{\log_2 N}{h} \leq R$. Choose N control levels: $u_k = -1 + \frac{2(k-1)}{N-1}$, $k = 1, \dots, N$. We claim that the interval $[-b/a, b/a]$ is invariant under the evolution of $x(j+1) = \alpha x(j) + \beta u(j)$ where the feedback law is defined by the selection function

$$u(j) = \begin{cases} 1 & \text{if } x(j) \leq -\frac{b}{a} + \frac{2}{N} \frac{b}{a} \\ 1 - \frac{2}{N-1} & \text{if } -\frac{b}{a} + \frac{2}{N} \frac{b}{a} < x(j) \leq -\frac{b}{a} + \frac{4}{N} \frac{b}{a} \\ \vdots & \vdots \\ 1 - \frac{2(N-2)}{N-1} & \text{if } -\frac{b}{a} + \frac{2(N-2)}{N} \frac{b}{a} < x(j) \leq \\ & -\frac{b}{a} + \frac{2(N-1)}{N} \frac{b}{a} \\ -1 & \text{if } -\frac{b}{a} + \frac{2(N-1)}{N} \frac{b}{a} \leq x(j) \end{cases} \quad (7)$$

With the feedback selection function defined in this way, we claim the closed-loop transition function $x(j+1) = \alpha x(j) + \beta u(j)$ leaves

the interval $[-b/a, b/a]$ invariant. This can be shown by showing that each subinterval

$$\left[-\frac{b}{a} + \frac{2(k-1)}{N} \frac{b}{a}, -\frac{b}{a} + \frac{2k}{N} \frac{b}{a} \right]$$

is mapped into the interval $[-b/a, b/a]$ precisely when $\alpha \leq N$, which will be true when the data-rate inequality is satisfied and we chose h and N as above. This proves the theorem. $\square$

3 The case of severely constrained feedback channels: Operating at the data-rate limit in the scalar case

For discrete systems (4) which arise from sampling continuous-time systems at a uniform rate, one can draw additional conclusions from the proof of Theorem 1. For instance, if the data-rate inequality is in fact an equality, and if the cardinality N of the set of admissible control values and the sampling interval h are chosen so that

$$a \log_2 e = \frac{\log_2 N}{h} = R,$$

then the control levels must be uniformly spaced. (See Figure 2.) Specifically, letting $u_{\min}$, $u_{\max}$ be the minimal and maximal elements of U and assuming without loss of generality that $u_{\min} = -u_{\max}$, the admissible control values may be explicitly enumerated

$$u_k = u_{\min} + \frac{2(k-1)}{N-1} \Delta$$

where $\Delta = u_{\max}$. Moreover, the (only) invariant interval is $[-\eta, \eta]$, where

$$\eta = \frac{\beta}{\alpha - 1} u_{\max} = \frac{b}{a} u_{\max}.$$

If we take $u_{\max} = 1$ (again without loss of generality), the (only) feedback law under which this interval is invariant and under which the motion of (4) will remain bounded is given by (7). Hence we have the following:

Corollary 1 Suppose the data-rate inequality holds with equality, i.e.

$$R = a \log_2 e.$$

If the discrete-time system (4) is obtained from a continuous-time system $\dot{x} = ax + bu$ by sampling at a uniform rate with sampling interval h , then bounded motions of (4) can be produced by a finite control set U only if

- (i) the cardinality N of U is such that $\log_2 N = Rh$,
- (ii) the only interval which is invariant under (4) using this control set is $[-(b/a)u_{\min}, (b/a)u_{\max}]$, and
- (iii) the selection function f (given by (7)) is unique (up to values on a set of measure 0), and $f^{-1}(U)$ defines a uniform partition of the interval $(b/a)u_{\min} \leq x \leq (b/a)u_{\max}$.

Additional information about the qualitative dynamics of (4) at the extreme data-rate limit is provided by the following:

Corollary 2 In the limiting case in which the data-rate inequality is satisfied with equality

$$R = a \log_2 e,$$

the trajectories of (4) are either bounded or unbounded depending on whether the initial value of the trajectory is in $[-(b/a)u_{\min}, (b/a)u_{\max}]$. If the initial value $x(0) \notin [-(b/a)u_{\min}, (b/a)u_{\max}]$, the resulting trajectory is not bounded. If $x(0) \in [-(b/a)u_{\min}, (b/a)u_{\max}]$, then the resulting trajectory remains confined to $[-(b/a)u_{\min}, (b/a)u_{\max}]$, and for almost all initial conditions in the interval, the trajectory is characterized statistically by the uniform probability density on the interval so that for every $-(b/a)u_{\min} \leq \varepsilon_1 \leq \varepsilon_2 \leq (b/a)u_{\max}$, the fraction of time the trajectory spends in the subinterval $[\varepsilon_1, \varepsilon_2]$ is

$$\frac{\varepsilon_2 - \varepsilon_1}{2(a/b)(u_{\max} - u_{\min})}.$$

Proof Again, without loss of generality, we may assume that $u_{\min} = -u_{\max}$, $u_{\max} = 1$, and the set of admissible control values is enumerated as in (7). Following the arguments in [2] and [4] we claim that the corresponding closed-loop transition function is ergodic on the invariant interval $[-b/a, b/a]$, and that its motions are characterized statistically by an invariant density which satisfies the functional equation

$$\rho(y) = \frac{1}{\alpha} \sum_{k=1}^N \rho\left(\frac{y - \beta(N+1-2k)/(N-1)}{\alpha}\right).$$

Up to a scalar multiple of its value there is a unique constant solution to this equation on $[-b/a, b/a]$ given by $\rho \equiv 1$. Hence, if $g(\cdot)$ is any Lebesgue measurable function on $[-b/a, b/a]$, and if we denote the closed-loop transition function defined by (7) by $F(\cdot)$, then by the Birkhoff ergodic theorem, for almost all $x \in [-b/a, b/a]$,

$$\lim_{k \rightarrow \infty} \sum_{j=0}^{k-1} g(F^j(x)) = \int_{-b/a}^{b/a} g(s) ds.$$

This proves Corollary 2. $\square$

Remarks on control at the data-rate limit: Although it may be possible to implement stable feedback control near the data-rate limit, no actual physical implementation of a stabilizing feedback law could be made to work exactly at the data-rate limit. Nevertheless, some remarks about this singular case will shed light on what may be expected of feedback and control coding designs which operate near the data-rate limit.

- (i) For given $R = a \log_2 e$, condition (i) of Corollary 1 implies that admissible sampling rates $(1/h)$ for systems operating at the data-rate limit are quantized: $h_k = (\log_2 k)/R$, $k = 1, 2, \dots$. In practice, such a condition could not be satisfied exactly—which is yet another reason that stable operation at the data-rate limit is not possible.
- (ii) Under our assumption of uniform sampling in time, we have shown that uniformity in quantization in the spatial domain and in the set of admissible control values is a consequence for feedback systems operating at the data-rate limit. Thus although nonuniform (e.g. logarithmic) spacing of control values in U may be of interest in some applications, we find that for quantizations which are near the critical case—cardinality $|U| \sim e^{ah}$ —we are forced to take the control values to be more or less evenly spaced. (A slightly different way to say this is that nonuniformly spaced control values require control sets of larger cardinality and higher data rates to stabilize a given open loop unstable system.)

- (iii) Analysis similar to the above shows it is possible to achieve stable operation of systems of the form (1) with sampling which is nonuniform in time. It is further possible to maintain stable operation even if the data rate occasionally falls below the data-rate limit of Theorem 1, provided the sampling rate is appropriately adjusted in terms of the system state and provided further that the averaged data-rate (over an extended time interval) satisfies the inequality (5). This suggests that it may be useful to develop state-dependent adaptive sampling strategies where the sampling rate is adjusted depending on the measured state of the system—say increasing the sampling rate depending on how far the system has moved from a desired operating point. This will be discussed elsewhere, and we refer in particular to [1] for some very recent work on novel sampling strategies. $\square$

4 How classical feedback designs divide controller attention in quantized control of higher order systems

The issues associated with limited data rates and coarse quantization are well illustrated by the scalar case treated in the previous sections. Binary control (i.e. cardinality $U = 2$) represents the coarsest possible quantization of the control values, and it remains an open question as to whether being within the data rate bounds that have appeared in the literature is sufficient to produce adequately regulated closed-loop dynamics in systems operating near the data-rate limit. To investigate these issues, we may rewrite (2) explicitly as

$$\begin{pmatrix} x_1(j+1) \\ x_2(j+1) \\ \vdots \\ x_n(j+1) \end{pmatrix} = \begin{pmatrix} \alpha_1 & 0 & \dots & 0 \\ 0 & \alpha_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \alpha_n \end{pmatrix} \begin{pmatrix} x_1(j) \\ x_2(j) \\ \vdots \\ x_n(j) \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} u(j). \quad (8)$$

We assume $\alpha_i > 0$ for all i , and $u(j) \in \{-1, 1\}$ for $j = 1, 2, \dots$

Remark 4 In equation (8), the i -th mode is the open-loop unstable if (and only if) $|\alpha_i| > 1$. Suppose $\alpha_1 > 1$, while for $i = 2, \dots, n$, $0 \leq \alpha_i < 1$. Then it is not difficult to show that the binary feedback law

$$u(x) = \begin{cases} 1 & \text{if } x_1 < 0 \\ -1 & \text{if } x_1 \geq 0 \end{cases} \quad (9)$$

produces bounded trajectories provided that $x_1(0) \in (-1/(\alpha_1 - 1), 1/(\alpha_1 - 1))$. Indeed, this feedback law stabilizes the unstable mode using the intuitively appealing rule which says that if the state x_1 is negative, the control value $u = 1$ should be chosen so as to increase the value of x_1 , while if $x_1 \geq 0$, the control value $u = -1$ should be chosen so as to decrease x_1 . All other modes (state components $x_2, \dots, x_n$) can be shown to remain bounded regardless of how we switch control values so as to bound x_1 . If there are two unstable modes, say $\alpha_1 > 1$ and $\alpha_2 > 1$, then the feedback law (9) will not typically keep the second

component $x_2(j)$ bounded. The challenge in control design in this setting is to choose binary sequences which will effectively resolve the conflicts that arise in trying to bound the two states simultaneously. In [4] it was shown that for systems (8), binary realizations of classical feedback laws could in fact produce bounded trajectories in the case in which there is more than one open-loop unstable mode provided that (8) is controllable. There is no chance of resolving the conflict between competing unstable modes if (8) is not controllable.

How quantized feedback divides its attention among system components. In [5], Brockett introduced the notion of *attention* in control. In [4], the question of how a quantized feedback control law divides its attention among the *modes* or components of (8) has been addressed. In considering the problem of using feedback to produce bounded trajectories of (8), the most interesting case occurs when the open-loop modes are all unstable—i.e. $\alpha_i > 1$. Consider feedback designs using binary control $U = \{-1, 1\}$. To keep the i -th component of the state bounded, the ideal choice of control value at the j -th time step will be $u(j) = -1$ if $x_i(j) \geq 0$ and $u(j) = +1$ if $x_i(j) < 0$. To discuss the selection function's choice of control value for each state x , label the orthants of $\mathbb{R}^n$ by means of n -vectors with components $+1$ and -1 , where the sign of the i -th component is the same as the sign of the component x_i for points in that orthant. Under this labeling convention, points lying on boundaries between orthants may be arbitrarily assigned membership in any orthant to which they are adjacent. In every orthant except those labeled either with all $+1$'s or all -1 's, however, any feedback law must resolve conflicting objectives of keeping the different component modes of the system bounded.

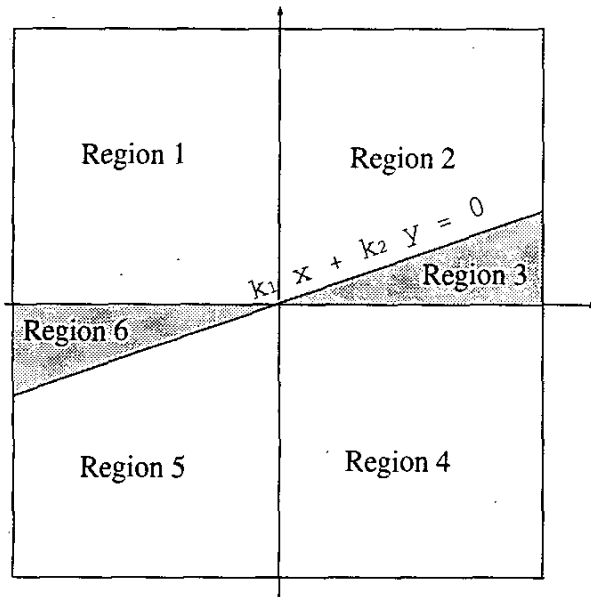


Figure 3: This figure illustrates how the quantized (binary) realization of the feedback law $u(x) = k \cdot x$ divides its attention between the two modes of the system. The selection function as defined in the text takes the value -1 above the line $k_1x + k_2y = 0$ and $+1$ on or below the line.

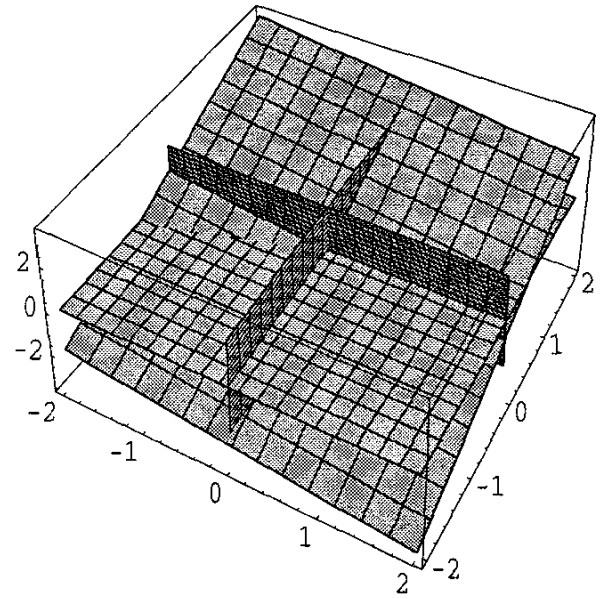


Figure 4: This figure is the three dimensional analogue of Figure 2, illustrating the way in which the binary realization of the feedback law described in Example 3 divides its *attention* among modes of the system. The skew plane is the locus of $k_1x_1 + k_2x_2 + k_3x_3 = 0$, and the feedback selection function takes the value -1 for points above this plane and the value $+1$ for points on or below the plane.

A remarkable and somewhat counterintuitive fact which emerges from the careful analysis of 2 and 3 dimensional systems below is that the binary valued selection functions associated with classical constant gain feedback designs move all state components in destabilizing directions in parts of the totally positive (labeled with all $+1$'s) and totally negative (labeled with all -1 's) orthants. To see this, we begin with a precise statement of what we mean by a binary realization of a constant gain feedback law. If the set of admissible control values U were an open subset of $\mathbb{R}$ containing the point 0, classical techniques would specify choices of constant gains $k \in \mathbb{R}^n$ such that with

$$u = u(x) = k \cdot x = k_1x_1 + \dots + k_nx_n,$$

the closed-loop system corresponding to (8) has $x = 0$ as an asymptotically stable rest point. The binary control which approximates this constant gain feedback is

$$u(x) = \begin{cases} -1 & \text{if } k_1x_1 + \dots + k_nx_n < 0 \\ 1 & \text{if } k_1x_1 + \dots + k_nx_n \geq 0. \end{cases} \quad (10)$$

Example 2 - Binary feedback control of a 2-d system. For a 2-d system, the binary control which approximates the constant gain feedback is

$$u(x, y) = \begin{cases} -1 & \text{if } k_1x + k_2y < 0 \\ 1 & \text{if } k_1x + k_2y \geq 0. \end{cases} \quad (11)$$

Suppose $\alpha_2 > \alpha_1 \geq 1$. Letting $u = k \cdot x = k_1x + k_2y$ be a constant-gain feedback law placing the closed-loop poles of (8) inside the

unit circle, we must have $-k_2 > k_1 > 0$. The above binary realization of this law partitions the 2-dimensional state space into the six regions depicted in Figure 3. The switching rule given by u may be described in terms of these regions as follows: If the state $(x(j), y(j))$ lies in either Region 2 or Region 5, the state transition $(x(j), y(j)) \mapsto (x(j+1), y(j+1))$ prescribed by u applied to (8) coincides with what it would be if the two states x and y were independently controlled using the canonical scalar binary control law (9). When $(x(j), y(j)) \in$ Region 1 or Region 4, the control law u favors the state y in the sense that the transition $y(j) \mapsto y(j+1)$ is prescribed by the canonical binary law

$$y(j+1) = \begin{cases} \alpha_2 y(j) + 1 & \text{if } y(j) < 0 \\ \alpha_2 y(j) - 1 & \text{if } y(j) \geq 0, \end{cases} \quad (12)$$

while the state x moves in the destabilizing direction. (I.e. $x(j) \mapsto \alpha_1 x(j) - 1$ if $x(j) < 0$ and $x(j) \mapsto \alpha_1 x(j) + 1$ if $x(j) \geq 0$.) Since the open-loop pole corresponding to the state y is farther from the unit circle than the pole for x , it is not surprising that the control law will favor a transition tending to stabilize y in Regions 1 and 4 where the objectives of stabilizing x and y are in conflict. It is somewhat counterintuitive that the feedback law never pays attention to the x -component in regions 1 and 4, because (as shown in [4]) in order for trajectories of (8) to remain bounded, the control law cannot always favor the y -component. It is precisely in Regions 3 and 6, where the absolute value of the x -component is considerably larger than the absolute value of the y -component, that the two open loop modes are treated equitably, and the control law relaxes its "preference" for keeping y bounded at the expense of x . It is somewhat surprising, however, that the way the control law pays attention to x in these regions is in making the transition prescribed by (11) which moves both components in the destabilizing direction. (I.e., for both x and y in Regions 3 or 6, the binary feedback law prescribes motion in the destabilizing direction: $x \mapsto x + \text{sgn}(x)$, $y \mapsto y + \text{sgn}(y)$.)

Example 3 - Binary feedback control of a 3-d system. For the case $n = 3$ and assuming $\alpha_3 > \alpha_2 > \alpha_1 > 1$, the operation of the binary selection function (10) associated with a standard stabilizing feedback law $u(x_1, x_2, x_3) = k_1 x_1 + k_2 x_2 + k_3 x_3$ is depicted in Figure 4. Again, the closed loop dynamics have a number of counterintuitive features. Examining Figure 4, it appears that the selection function subdivides each orthant except the (symmetrically located) two labeled $(1, -1, 1)$ and $(-1, 1, -1)$ (in accordance with our labeling procedure). For all points in these two regions, the binary feedback (10) law favors keeping x_1 and x_3 as close to 0 as possible—always at the expense of moving x_2 farther away from 0. Furthermore, in the totally positive, $(1, 1, 1)$ (resp. totally negative, $-1, -1, -1$) orthants, where the intuitive choice of input is unambiguously $u = -1$ (resp. $u = +1$), we find that the selection function (10) prescribes motion tending to move all components away from 0 when $|k_3 x_3(j)| < |k_1 x_1 + k_2 x_2|$.

5 Operating near the data-rate limit in higher dimensions

The calculations of the preceding sections underscore the fact that asymptotic stability is only an abstraction which is not achievable in practice using finitely quantized control. Thus it is of interest (particularly in the case of severe data-rate constraints) to describe the ω -limit sets of various quantized feedback designs. That LTI systems with quantized feedback can typically be locally but not globally stabilized is made precise in the following proposition.

Proposition 1 Local stabilization by quantized feedback. *For any feedback law implemented using quantized control with a finite set U of control values, the closed loop system is only locally stable in the sense that there is a compact set such that any trajectory of (3) which leaves this set must be unbounded.*

Proof A version of this proposition is proved in [4]. $\square$

One is thus interested in bounded attractors and their corresponding regions of attraction. For the scalar case treated in Section 3, the results are highly explicit and suggest what is to be expected in higher dimensions. As the data rate decreases to its theoretical minimum, the largest attracting set for the closed-loop system collapses onto the attractor. At the data-rate limit, it follows from Corollary 2 that the closed-loop dynamics are chaotic and ergodic within the attractor. Thus, in the data-rate limit, we can see the exact mechanism by which stability is lost in the scalar case.

In the multidimensional case, the data-rate inequality is

$$R \geq (\mathcal{R}(\lambda_1(A)) + \dots + \mathcal{R}(\lambda_k(A))) \log_2 e,$$

where $\lambda_1(A), \dots, \lambda_k(A)$ are the open-loop eigenvalues of (1) having positive real parts. (C.f. [10].) Consider (8) in the case $n = 2$. We shall examine cases in which both $\alpha_1 > 1$ and $\alpha_2 > 1$. By way of analogy with the scalar case, we would expect this system to be boundable using the binary feedback law (11) precisely when $\alpha_1 \alpha_2 \leq 2$. Calculations to date indicate, however, that stable behavior near the data-rate limit is much more fragile in the case of dimensions 2 and higher.

Figure 5 shows attracting sets for three cases in which $\alpha_1 \alpha_2 \leq (1 + \sqrt{5})/2 = 1.6180339887498948482\dots$. These have been generated by simulating binary implementations (11) of a feedback law designed along classical lines to place the closed-loop poles at $(0.5, 0.5)$. In all cases, the domains of attraction are not significantly larger than the attractors themselves. The attractor in Figure 5(c) may be explicitly and exactly described. (See [4].) The figure actually consists of five vertical lines, symmetrically arranged with respect to the origin. The motions of (8) confined to this attractor are ergodic on all five vertical strips, and in accordance with a probabilistic transition law determined again from the ergodic theory of mappings, trajectories jump from one strip to another so that almost all trajectories visit all five strips. Based on simulations, the set of points attracted to this set is extremely thin, indicating that this system is effectively at the data-rate limit.

6 Conclusion

This paper has examined the behavior of LTI feedback systems operating under severe communications constraints on feedback loops. In the case of scalar systems, very explicit calculations show exactly how systems lose stability under severe data-rate constraints. For systems of higher dimension, we have examined limited data-rate control in terms of Brockett's concept of *attention*. The behavior of the coarsest possible (binary) quantization of classical feedback laws has been examined in the case of two and three dimensional examples. Again, the way in which stable operation degrades in the face of severe data-rate constraints has been observed. The results of this paper point fairly clearly to the conclusion that it is very difficult to operate control systems that

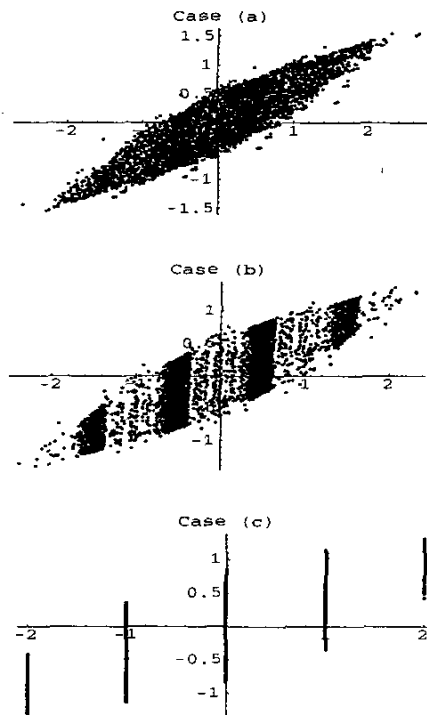


Figure 5: Simulations indicate the appearance of attracting sets for the discrete-time system (8) under the binary feedback law (11). The cases are: (a) $(\alpha_1, \alpha_2) = (1.05, 1.5)$, (b) $(\alpha_1, \alpha_2) = (1.01, 1.55)$, and (c) $(\alpha_1, \alpha_2) = (1, (1 + \sqrt{5})/2)$.

have hard real-time constraints (e.g. right half-plane poles) near the theoretical data-rate limit. Nevertheless, if the communication constraints are not extremely severe, there may be effective coding strategies which will support acceptable performance of the closed-loop system. As the author was finishing the final version of this paper, the Proceedings of 2002 MTNS Conference became available, and interesting work related to such coding questions has appeared in [6] and [7].

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